

1.3 FBLA STUDENT ACTIVITIES

IST/CSP

Micro:bit Smart Hall Pass System

Instructions:

Assignment Overview

Your team has been hired by Centennial High School to design a **Smart Hall Pass System** using Micro:bits.

The goal is to replace a traditional paper hall pass with a digital wireless communication system.

Your final system must allow a student to:

- Request a hall pass
- Select the type of pass needed
- Send the request wirelessly
- Receive approval or denial from the teacher

Your final system must allow a teacher to:

- Receive requests
- View Student ID
- View Pass Type
- Approved requests
- Deny requests
- Track requests using enhancements

Team Structure

Student Device Team (2 Students)

Responsible for:

- Student ID
- Pass Type Selection
- Request Transmission
- Approval/Denial Displays

Teacher Device Team (2 Students)

Responsible for:

- Request Reception
- Approval System
- Statistics
- Dashboard Features

Timeline

Tuesday

Project Introduction and Starter Code

Wednesday

Student IDs and Pass Types

Thursday

Two-Way Communication

Friday

Enhancements and Final Testing

Monday

Whole-Class Live Demonstration

FBLA Event: Sales Presentation

Students design and build a Micro:bit-powered hall pass system intended to streamline student movement tracking and improve school operations. Students create a prototype, prepare a product pitch, and present the solution as a technology product.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback

The screenshot displays a Microsoft Teams interface. At the top, a header bar shows the assignment title 'MJ2 - Hall Pass', the due date 'Due: Sep 14 at 11:59 PM', and the submission status '46/46 Graded', '89.57 / 100 (90%) Average', and '30/46 Students'. Below this, a toolbar includes icons for zooming and other editing functions. The main content area shows a student's submission of a Micro:bit program. The program is written in a block-based language and includes several radio communication blocks. On the right side of the interface, there is an 'Assessment' section showing a grade of '100%' and a status of 'None'. Below the assessment, there are 'Assignment Comments' from a teacher named Dina Madaoui, dated 'Sep 14 at 8:35 PM'. The comments include a positive feedback message and a question about the student's experience with the project.

AP Computer Science Principles

What's the Odds? Powerball vs. Bitcoin Mining

Instructions:

Scenario

A software company wants to create an application that compares two possible ways of making money with an initial investment of \$1,000:

- Buying Powerball tickets
- Mining Bitcoin

Before programmers write code, they need researchers to determine:

- What information is required
- What assumptions must be made
- What mathematical concepts are needed
- What variables should be included
- What errors could occur

Your role is to act as a member of the software company's research team.

You are not coding during this phase of the project.

Driving Question

If you had \$1,000 and a two-year time period, would you have a better chance of making money from:

- Powerball tickets
- Bitcoin mining

Support your conclusion using research and evidence.

Learning Objectives

Big Idea 1.1 Collaboration

- Work with classmates to discuss findings and ideas.
- Compare sources and information.
- Ask questions and provide feedback during the gallery walk.

Big Idea 1.2 Program Function and Purpose

- Determine what problem the future software should solve.
- Identify the purpose of the simulation.
- Determine what information users would want from the program.

Big Idea 1.3 Program Design and Development

- Identify variables needed before coding begins.

- Research required data sources.
- Determine what calculations and inputs the software would need.

Big Idea 1.4 Identifying and Correcting Errors

- Identify assumptions that could lead to inaccurate results.
- Determine possible mistakes programmers could make.
- Explain how those mistakes could be prevented.

FBLA Event: Computer Problem Solving

Students investigated and compared the probability of winning the Powerball lottery versus successfully mining Bitcoin. Students researched the problem, gathered supporting information, analyzed the mathematics required for comparison, explored probability concepts, and followed a design-thinking process to develop a proposed software solution before implementation.

Evidence Required

- One Graded Student Sample
- Teacher Feedback

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ZOOM +

Powerball research

- According to Powerball itself (powerball.com) Each ticket costs 2 dollars
- Doing the math, if you spent the whole 1000\$ on tickets, it would total to 500 tickets
- In order to play powerball, you'd pick 5 numbers from 1-69, as well as a Powerball number from 1-26
- Even though the amount of tickets may sound like we have good odds, the projected jackpot odds are very slim. Doing this math, this is the amount of possible combinations from 5 numbers between 1-69

$$\left(\frac{69}{5} \right) = \frac{69!}{5!(69-5)!} = 11,238,513$$

- Afterwards, we must multiply this by 26, which in total gives us 292,201,338, which means our final odds to win the jackpot prize in the powerball to 1 in 292201338. This is helped by our purchasing of 500 tickets, where it is lessened to 1 in 584,403.
- As well as this, we are almost guaranteed to be able to get more money from smaller prizes, but since it all falls on chance, we ultimately cannot bet on making a profit, let alone making it big

Submission to view:

Aug 20 at 12:02 AM

Word Count: 577

Submitted Files: (click to load)

Bitcoin Vs Powerball by Ismail2.pptx

Assessment

Grade out of 100

99

Status

None

Assignment Comments

Great use of resources and also good analytical skills to identify the right probability model, including math equations to solve the two models before you are able to compare them and generate a solid conclusion of the odds.

AP Computer Science A

Technology Leadership Speaking Challenge

FBLA Event: Impromptu Speaking

Students receive a technology topic, are given twenty minutes to prepare, and deliver a five-to-seven-minute speech demonstrating communication, leadership, and technical understanding.

Evidence Required

- One Graded Student Sample
- Teacher Feedback

Game Design and Simulation

Environment and Emotional Design Analysis

Students evaluate how environmental design, audio, storytelling, color theory, level design, and gameplay mechanics affect player engagement and overall user experience.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback

The image shows a presentation slide titled "Mechanics" on a red background. The slide contains two text boxes and a screenshot of a game environment. The first text box states: "The player can move around the place, and interact with the robot citizens there. As you talk to the robots, they give you informations, clues, and even details about their daily life. This allow you to both find clues for your mission and immerse in the environment." The second text box contains a screenshot of a game environment showing a character in a green and white striped shirt interacting with a robot in a room with plants and a window. The right-hand sidebar displays assignment comments and a rubric. The comments include: "i enjoyed it, thanks for sharing" and "As you went through observing and analyzing the gameplay you can realize the importance of the ambient, the environment, every detail counts for capturing the player, to help with the engagement with music, animations, objects, colors that the game intent to allow the player to follow the story through exploration." The rubric is titled "Rubric: All completed" and lists criteria: "Explain location", "use of elements", "Mechanics", "Level of Intensity", "How to spark Intensity", "what creates emotions", and "thoughts about the overall story".